

Task 1 Report: Build & Evaluate a Linear Regression Model (House Price Predictor)

Objective

Train and evaluate a Linear Regression model using the California Housing dataset. Perform exploratory data analysis (EDA), split the data into training and testing sets, evaluate model performance, and visualize the results.

Dataset

California Housing dataset from scikit-learn. Target variable: MedHouseVal (Median House Value). Features include Median Income, House Age, Average Rooms, Average Bedrooms, Population, Average Occupancy, Latitude, and Longitude.

Tools & Libraries

Python, Pandas, NumPy, Matplotlib, Seaborn, Scikit-learn, Jupyter Notebook.

Methodology

1. Imported required libraries.
2. Loaded the California Housing dataset.
3. Performed EDA using `head()`, `info()`, `describe()`, missing value check, correlation heatmap, and histograms.
4. Split data into training and testing sets (80:20).
5. Trained a LinearRegression model.
6. Predicted house prices on the test set.
7. Evaluated the model using MAE, RMSE, and R^2 Score.
8. Plotted Actual vs Predicted and Residual Plot.
9. Saved the trained model as `house_price_model.pkl`.

Observations

- Median Income showed the strongest positive relationship with house price.
- No missing values were found in the dataset.
- The model produced reasonable predictions for most houses, though errors increase for very high-priced houses.

Conclusion

The Linear Regression model successfully learned relationships between housing features and house prices. It provides a good baseline regression model and demonstrates the complete machine learning workflow from data loading to evaluation.

Future Improvements

- Perform feature engineering.
- Tune hyperparameters.
- Compare with Decision Tree, Random Forest, or Gradient Boosting models.
- Build a simple prediction interface using Streamlit or Flask.